Occlusion-Aware Bilayer Modeling + SAM: Deep Theoretical and Experimental Review

This document reviews only the scientific method and reported experiments of the 2025 occlusion-aware bilayer paper. Repository ancestry, source files, exact released configuration, and execution details are separated into the companion repository document.

1. Problem Formulation

Prohibited-item segmentation in X-ray baggage differs from natural-image segmentation in two important ways. First, material-dependent X-ray absorption produces colors and textures that do not match ordinary photographs. Second, objects are projected through one another, producing severe overlap and weak boundaries. A detector can localize the correct item but merge its mask with another item or surrounding baggage.

The paper proposes:

- a Mask R-CNN-style automatic instance-segmentation pipeline using SAM ViT-L features and SAM mask decoding;
- a learned RoI-to-prompt mechanism rather than interactive user prompts;
- two sequential SAM mask decoders: one for the projected occluder region and one for the target item;
- residual guidance from the first decoder's transformer-refined image/prompt embeddings to the second;
- PIDray-A and PIXray-A, which add manually annotated overlap regions to existing X-ray datasets;
- controlled comparison with same-backbone RSPrompter and component ablations.

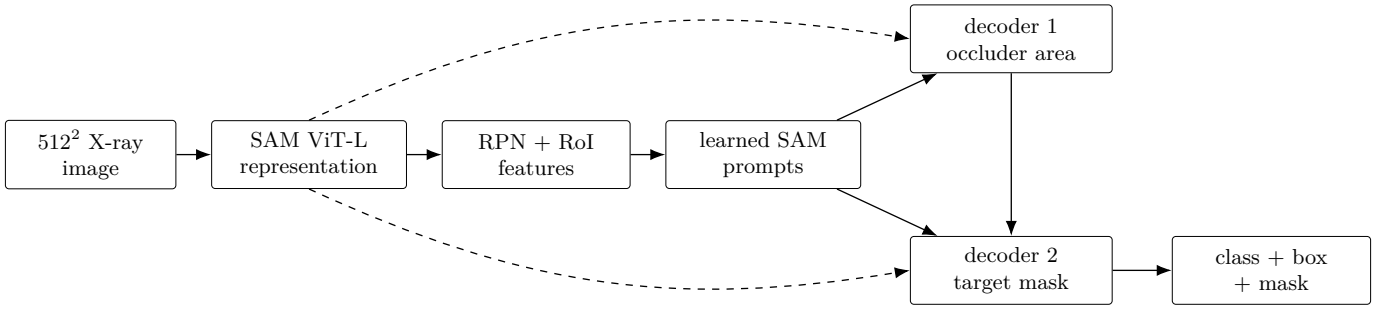


Figure 1: Scientific pipeline, independently reconstructed.

2. Occlusion Annotation Semantics

For image I, target instance k has class c_k, box b_k, item mask M_k, and an extra occluder-region mask O_k. O_k represents the projected mask area of other prohibited objects overlapping the target. It is not an amodal target completion, physical depth order, or generic background mask.

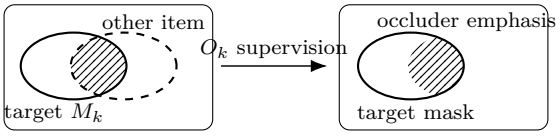


Figure 2: The extra supervised region used by the first decoder.

The paper identifies targets whose masks are covered by other-object boxes over a threshold, verifies true mask intersection, and records the intersecting projected region. It states that the overlap areas were manually annotated and reorganized with the original PIDray/PIXray information.

Table 1: Published PIDray-A and PIXray-A statistics.

	PID train	PID Easy	PID Hard	PID Hidden	PIX train	PIX test
Total images	76,913	24,758	9,746	13,069	3,560	1,486
Positive images	29,454	9,482	3,733	5,005	3,560	1,486
Multi-item images	7,411	0	3,733	3	2,891	1,193
Occlusion images	4,785	0	2,434	0	1,550	654
Total instances	39,708	9,482	8,892	5,008	10,709	4,508
Extra annotations	11,080	0	5,585	0	4,114	1,765

PIDray-A retains normal images and the original Easy/Hard/Hidden protocol. PIXray contains only positive images and is split 80/20. PIDray Hard and a dedicated occlusion subset are the strongest mechanism tests because Easy and Hidden have no published extra overlap annotations.

3. Model Theory

3.1 SAM representation and automatic detection

The pipeline uses SAM ViT-L as the image representation. A feature pyramid supplies multiple resolutions to a standard RPN and RoI box head. This converts a foundation model designed for prompted segmentation into an automatic detector: RPN proposes regions, the box head predicts classes/locations, and positive RoI features become SAM-format prompts. For positive RoI feature $R \in \mathbb{R}^{256\times14\times14}$, a learned prompt head produces five 256-dimensional sparse tokens:

$$R \xrightarrow{\text{conv + MLP}} \mathbb{R}^{5\times(2\cdot256)} \xrightarrow{\text{sine/pair collapse}} S \in \mathbb{R}^{5\times256}.$$

A learned no-mask token is broadcast as the dense prompt. Full-image features, image position, and proposal-specific sparse prompts then enter the mask decoder.

3.2 Bilayer decoder

Let E be the image embedding, P its positional encoding, S the sparse RoI prompt, and D the dense no-mask prompt. The occluder decoder first predicts overlap mask $\widehat{O}$ and returns transformer-refined image/prompt states:

$$(\widehat{O}, \widehat{E}_o, \widehat{S}_o) = \mathcal{D}_o(E, P, S, D).$$

Residual guidance adds those refined states to the original states:

$$\widetilde{E} = (E + D) + \widehat{E}_o, \qquad \widetilde{S} = S_{\text{with output tokens}} + \widehat{S}_o.$$

The target decoder predicts the prohibited-item mask:

$$\widehat{M} = \mathcal{D}_t(\widetilde{E}, P, \widetilde{S}, D).$$

The important conceptual point is that guidance uses the first decoder's optimized embeddings, not only its binary mask. Cross-attention has already emphasized overlap relationships before those features enter the target decoder.

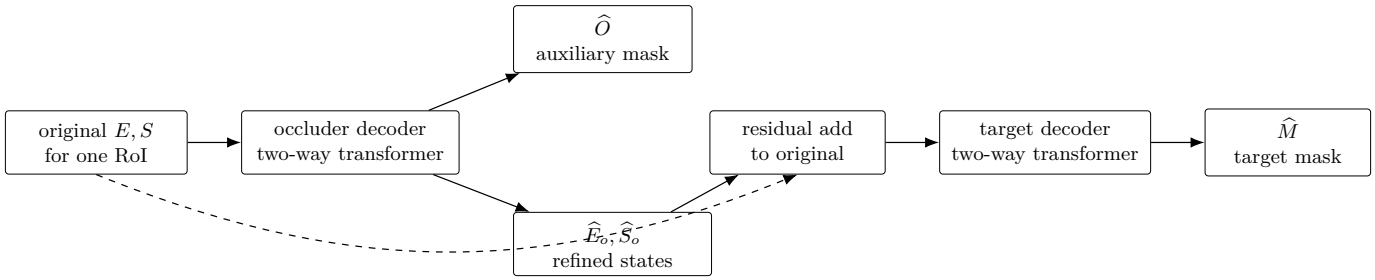


Figure 3: Bilayer decoder and residual guidance.

3.3 Multi-task objective

The model retains Mask R-CNN's RPN classification/regression and RoI class/box losses. The segmentation term becomes

$$L_{\text{seg}} = \text{BCE}(\widehat{M}, M) + \lambda_o \,\text{BCE}(\widehat{O}, O), \qquad \lambda_o = 0.25.$$

The total loss is

$$L = L_{\text{rpn}} + L_{\text{cls}} + L_{\text{box}} + L_{\text{seg}}.$$

The auxiliary term forces explicit overlap estimation; residual guidance determines whether that knowledge directly changes target-mask reasoning.

4. Experimental Protocol

Table 2: Paper-level experimental setup.

Element	Reported setup
Datasets	PIDray-A and PIXray-A with added occlusion supervision.
Backbone	SAM ViT-L; input resized to 512×512.
Optimization	AdamW; 50-iteration warm-up; cosine annealing.
Occluder weight	0.25.
Augmentation	Horizontal flip and random jitter.
Hardware/batch	One NVIDIA A800 80 GB; batch 64.
Epochs	50 on PIDray-A; 60 on PIXray-A.
Metrics	COCO box/mask AP, AP_{50}, and AP_{75}; subset evaluation.

Comparison methods are Mask R-CNN, Cascade Mask R-CNN, the PIDray journal model, SDANet for subset analysis, and RSPrompter with the same SAM backbone. The RSPrompter comparison is the cleanest test of the bilayer contribution because it controls the foundation-model representation and automatic prompt mechanism.

5. Published Overall Results

Table 3: Published overall results.

	PIDray-A						PIXray-A					
Model	B AP	B50	B75	M AP	M50	M75	B AP	B50	B75	M AP	M50	M75
Mask R-CNN	58.7	76.2	67.2	50.2	73.4	58.2	65.2	91.1	75.3	56.1	86.7	61.7
Cascade Mask R-CNN	65.0	78.1	72.1	53.6	76.0	62.4	71.8	91.2	81.0	57.1	86.2	63.1
PIDrayNet	66.6	81.7	74.3	53.4	78.6	61.4	72.4	91.6	81.3	57.7	87.1	64.1
RSPrompter	66.0	83.1	74.3	56.5	81.4	64.4	72.0	94.3	83.1	58.9	90.1	66.4
Bilayer	67.5	84.3	75.8	57.6	82.6	65.8	72.0	94.4	83.7	58.9	90.6	66.2

On PIDray-A, Bilayer improves same-backbone RSPrompter by +1.5 box AP and +1.1 mask AP. On PIXray-A, overall box and mask AP are unchanged, although threshold-specific results move slightly. The mechanism therefore helps clearly on PIDray-A but is not universally dominant.

6. Difficulty and Occlusion Results

Table 4: Published PIDray-A subset AP.

	Box AP			Mask AP		
Method	Easy	Hard	Hidden	Easy	Hard	Hidden
SDANet	72.5	63.7	48.0	61.1	51.7	37.0
PIDrayNet	74.5	64.8	53.0	61.4	51.9	39.7
RSPrompter	74.1	67.2	45.3	65.7	58.8	33.2
Bilayer	75.7	68.3	47.1	67.1	60.0	34.3

Bilayer improves RSPrompter on all six difficulty measures, especially Easy and Hard segmentation. It remains below PIDrayNet on Hidden boxes and masks. The paper attributes this to PIDrayNet's X-ray-specific hidden-object mechanisms and the SAM pipeline's simpler proposal stage.

Table 5: Published validation and occlusion-subset analysis.

Set	Model	B AP	B50	B75	M AP	M50	M75
Val	PIDrayNet	77.2	88.1	84.8	67.3	87.5	80.0
	RSPrompter	78.6	91.9	88.1	70.6	91.1	82.8
	Bilayer	78.7	92.0	88.2	70.7	91.4	83.1
Occ	PIDrayNet	49.8	64.0	55.2	37.8	60.7	40.3
	RSPrompter	46.7	67.7	54.4	39.4	63.0	42.5
	Bilayer	47.5	67.9	55.4	40.3	64.1	44.1

On occlusion images, Bilayer gains +0.8 box AP and +0.9 mask AP over RSPrompter. PIDrayNet retains higher box AP, while the SAM models give higher mask AP. This separates localization strength from segmentation quality.

7. Component Ablation

Table 6: Published occluder/guidance ablation.

Occluder	Guide	B AP	B50	B75	M AP	M50	M75
No	No	66.1	83.2	74.2	56.6	81.3	64.7
Yes	No	66.6	83.5	74.9	56.9	81.8	64.8
Yes	Yes	67.5	84.3	75.8	57.6	82.6	65.8

The auxiliary occluder decoder alone adds +0.5 box AP and +0.3 mask AP. Residual guidance adds a further +0.9 box AP and +0.7 mask AP. The larger second gain supports the intended mechanism: overlap information must influence target reasoning, not merely be predicted as an isolated auxiliary task.

8. Scientific Interpretation and Limitations

Table 7: Evidence boundaries.

Observation	Interpretation
Same-backbone RSPrompter gain	Strongest evidence that the bilayer module adds value.
PIDray-A gain, PIXray-A tie	Effect depends on dataset composition and overlap distribution.
Higher Hard masks	Explicit overlap modeling helps multi-item segmentation.
Weak Hidden results	Occlusion modeling does not replace hidden-item proposal/localization mechanisms.
Auxiliary + guidance ablation	Both supervision and feature transfer matter; guidance contributes more.
Manual overlap annotations	Extra supervision increases data cost and limits direct use on unannotated datasets.
No depth/amodal annotation	Claims should remain about projected overlap, not physical occlusion order.
Large SAM ViT-L backbone	Accuracy comparisons should include compute, memory and latency.

For the proposed research, Bilayer is the direct occlusion-aware baseline. It should be compared with RSPrompter under the same backbone, and with PIDray to distinguish X-ray-specific detection improvements from foundation-model segmentation improvements.

9. Primary Source

Ren, Y., Li, R., Liu, L., and Chen, C. (2025). *Prohibited Items Segmentation via Occlusion-aware Bilayer Modeling.* IEEE International Conference on Multimedia and Expo. arXiv:2506.11661.